Software Requirements Specification (SRS)

Defence-Style Embedded Mission Support and Communication Controller (MESCC)

Prototype / Academic Dataset — Synthetic and Non-Operational

Purpose: This document is a synthetic SRS created for developing and evaluating an AI/RAG-based requirement-to-test-case traceability and coverage analysis system. It is not a real defence system specification and contains no operational BEL information.

1. Scope

The system provides authenticated command/configuration access, subsystem health monitoring, communication-link management, message integrity checks, event logging, fault handling, watchdog recovery, time synchronization, and controlled restart recovery for an embedded mission-support controller.

2. Operating Assumptions

The controller operates on an embedded computing platform with a primary and backup communication interface, synchronized time source, persistent storage, hardware watchdog, and monitored power/processing resources.

3. Security and Safety

The prototype assumes least-privilege access, protected configuration parameters, auditable operator actions, safe-state behavior for critical faults, and rejection of malformed or integrity-invalid inputs.

4. Requirement Identification

Requirements use IDs REQ-001 through REQ-030. Each requirement is written as a testable 'shall' statement so that it can be mapped to verification test cases.

5. Verification Approach

Requirements are intended to be verified using functional, timing, fault-injection, recovery, security, integrity, and endurance-oriented tests. The accompanying test sheet intentionally includes both direct and partially redundant test cases to provide realistic retrieval candidates.

6. System Requirements

ID	Subsystem	Requirement
REQ-001	System Initialization	The system shall complete power-on self-test (POST) within 30 seconds and report the status of all monitored subsystems.
REQ-002	System Initialization	The system shall enter operational mode only when all mandatory subsystems pass POST.
REQ-003	Authentication	The system shall require operator authentication before allowing access to configuration functions.
REQ-004	Authentication	The system shall lock operator access for 5 minutes after three consecutive failed authentication attempts.
REQ-005	Command Interface	The system shall validate every operator command against the permitted command set before execution.
REQ-006	Command Interface	The system shall reject malformed or unsupported commands without changing system state.
REQ-007	Health Monitoring	The system shall monitor processor, memory, storage, communication, and power health at intervals not exceeding 5 seconds.
REQ-008	Health Monitoring	The system shall generate a health alert when a monitored parameter exceeds its configured threshold.
REQ-009	Communication	The system shall establish the primary communication link within 10 seconds after entering operational mode.
REQ-010	Communication	The system shall detect loss of the primary communication link within 3 seconds.
REQ-011	Communication	The system shall switch to the configured backup communication link within 10 seconds after primary-link failure.
REQ-012	Communication	The system shall maintain message sequence numbers and reject duplicate messages.
REQ-013	Data Integrity	The system shall verify the integrity of received messages using the configured checksum/CRC mechanism.
REQ-014	Data Integrity	The system shall discard messages that fail integrity verification and record an event in the system log.
REQ-015	Event Logging	The system shall timestamp security, communication, fault, and operator-command events with synchronized system time.
REQ-016	Event Logging	The system shall retain at least 30 days of event logs under nominal logging conditions.
REQ-017	Fault Handling	The system shall place a failed subsystem into a safe state when a critical fault is detected.
REQ-018	Fault Handling	The system shall generate a critical fault indication within 2 seconds of detecting a critical fault.
REQ-019	Watchdog	The system shall use a hardware watchdog to initiate recovery when the application becomes unresponsive for more than 5 seconds
REQ-020	Watchdog	The system shall record a watchdog reset reason after recovery.
REQ-021	Time Synchronization	The system shall synchronize its system clock with the configured time source when the source is available.
REQ-022	Time Synchronization	The system shall raise a time-synchronization fault when clock offset exceeds 100 milliseconds.
REQ-023	Configuration	The system shall prevent unauthorized modification of protected configuration parameters.
REQ-024	Configuration	The system shall validate configuration values against their permitted ranges before storing them.
REQ-025	Configuration	The system shall preserve the last valid configuration after an unsuccessful configuration update.
REQ-026	Security	The system shall terminate an authenticated configuration session after 10 minutes of operator inactivity.
REQ-027	Performance	The system shall process a valid incoming status message within 200 milliseconds under nominal load.
REQ-028	Availability	The system shall automatically restart the communication service after a recoverable service failure.
REQ-029	Audit	The system shall log successful and failed attempts to access protected configuration functions.
REQ-030	Recovery	The system shall restore essential operational services within 60 seconds after a controlled system restart.